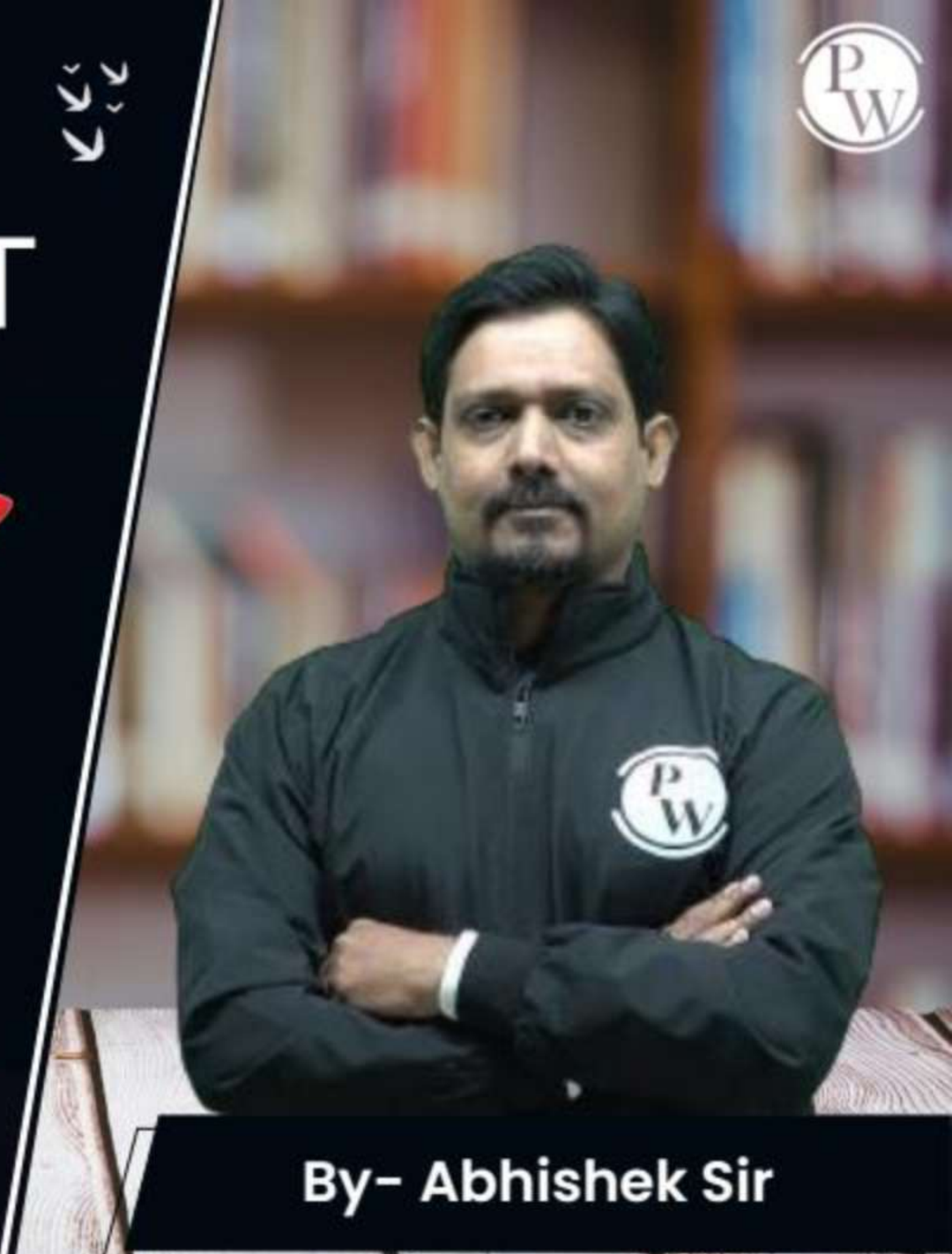
Computer Science & IT



C programming

Control Flow Statement

Lecture No. 05



By- Abhishek Sir

Recap of Previous Lecture



Topic while Loop

Topic do - while Loop

Topic

Topic

Topic

Slide

Topics to be Covered

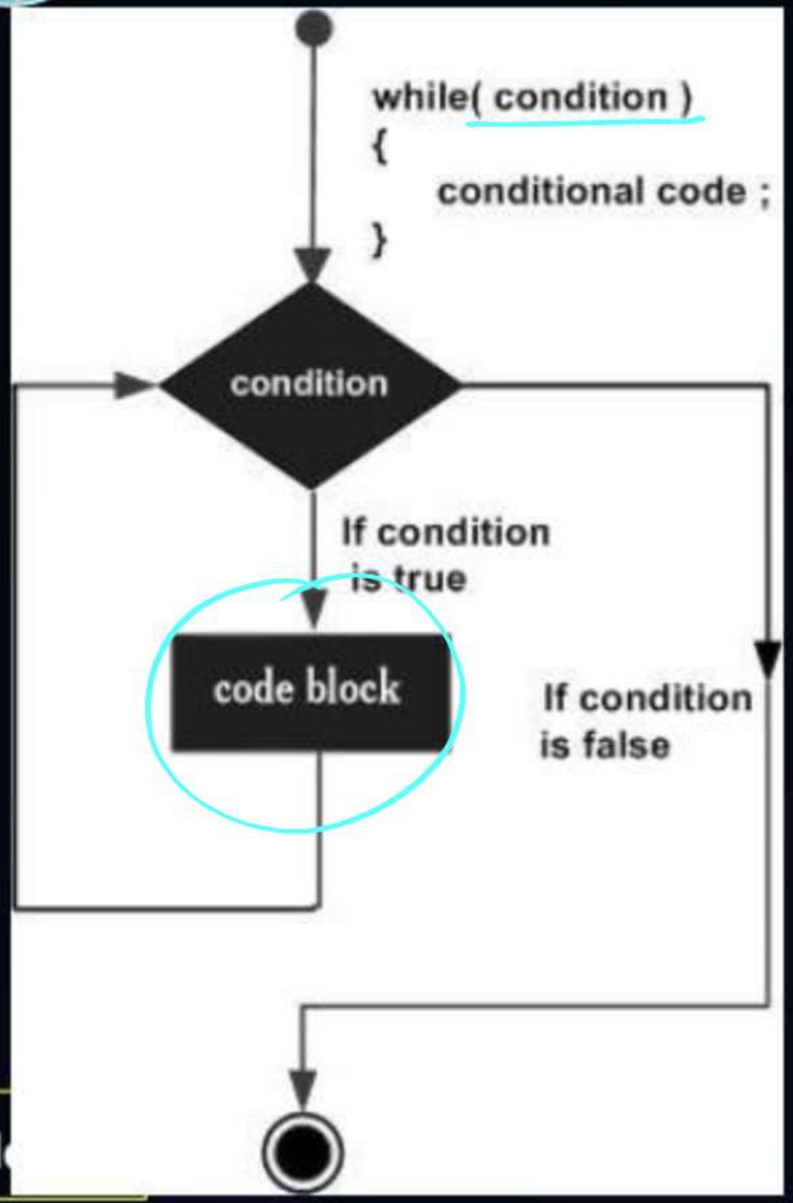


- Topic
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Continue
break
} Loop control statement.

Slide

while loop

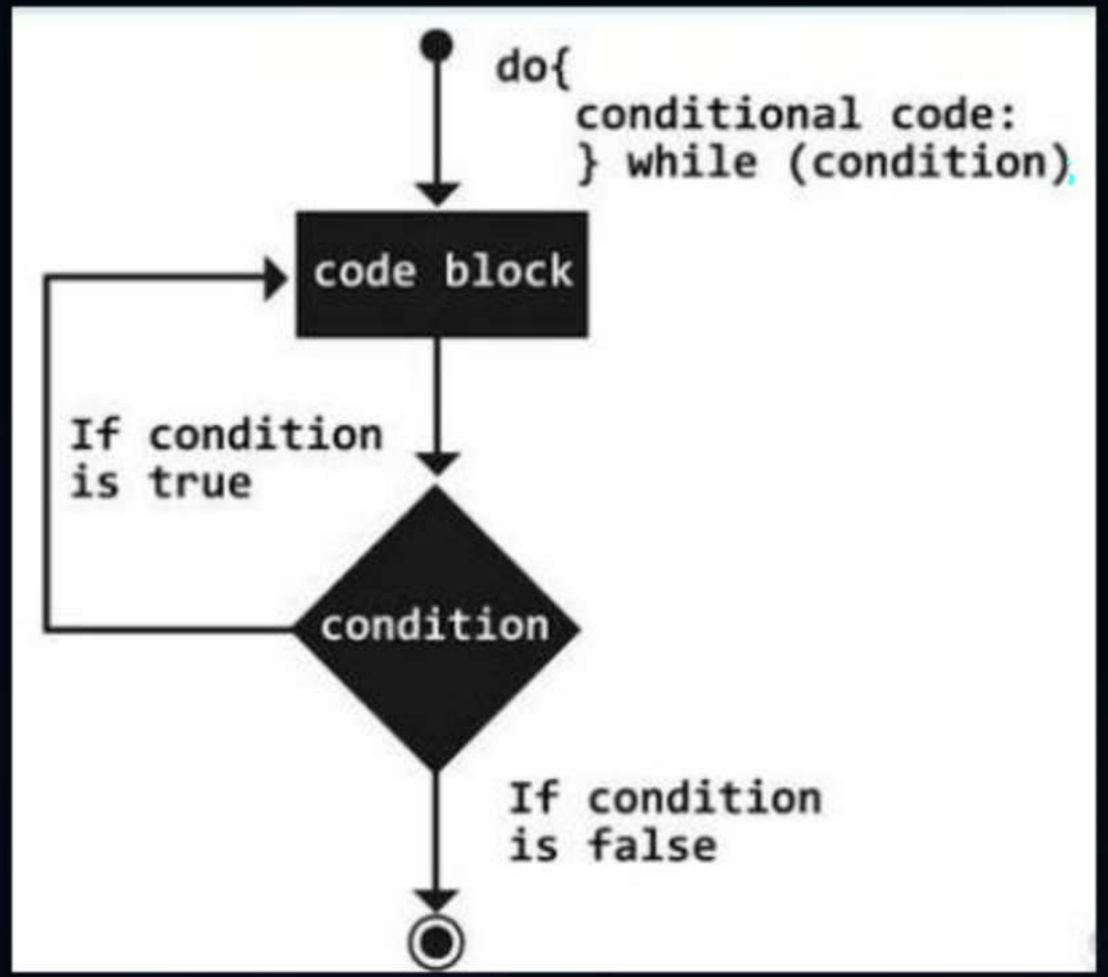


Linearly increament → for loop

Entry controlled



do-while loop



exit controlled

if condition is false then

1 time always executes

Slide



Question

Consider the following program segment

```
#include<stdio.h>
int main(){
      int x, i, j;
      x = 0;
      for (i = 0;i<20; i++)
            for(j=i+1;j<18;j++)
                  x++;
      }
```

The value of x after executing the segment

(A) 171 (B) 153 (C) 342 (D) 380

i:0 — 1 — 17 — 17
i:1 — 2 — 17 — 16
i:2 — 3 — 17 — 15
i:3

i:16 — 17 — 17 — 1
i:17 — 18
i:18
i:19

1+2+3+...+17

$\frac{17 \times 18}{2} = 153$

Slide



Question

Consider the following Program

```
#include<stdio.h>
    int main(){
        int i=1, j=1;
        for(; j; printf("%d%d\t",i,j))
            j=i++ <= 5;
        return 0 ;
    }
```

(a) 11 22 33 44 5

(b) 21 31 41 51 61

(c) 21 31 41 51 61 70 ✓

(d) Infinite loop

21, 31, 41, 51, 61, 70

i=1 j=1 j=1<=5 j=1, i=2

i=2 j=1 j=2<=5 , j=1, i=3

i=3, j=1 j=3<=5 , j=1, i=4

i=4 j=1 j=4<=5 , j=1 5

i=5 j=1 j=5<=5 j=1, i=6

i=6, j=1 j=6<=5 j=0 , i=7

Slide



Question

```
#include <stdio.h>
int main() {
    int i,j=10;
   for(i =1; i<=3; i++){
        do{
            printf("I am good Student");
            j = j+10;
            i++;
        }while(j<=200);
   }

    return 0 ;
}
```

The number of times printf executed is _____

$$\frac{200-10+1}{10} = \left\lceil \frac{191}{10} \right\rceil = 20 \text{ times}$$

i = 1 — 20 times

i = 20 — j = 210 + 10 = 220 — ①

i = 30 — j = 220 + 10 = 230 — ①

j : 20 30, 40, 50, 60, 70, 80, 90

100 110 120 130, 140, 150 160, 170, 180

190 200 210

22

Slide



Question

PYQ

Consider the following C code. Assume that unsigned long int type length is 64 bits.

```
unsigned long int fun(unsigned long int n){
      unsigned long int i, j=0, sum = 0;
       for( i=n; i>1; i=i/2)
              j++;
      for( ; j>1; j=j/2)
              sum++;
      return sum;
}
```

n = 2^{40}

j value will be?

$\lceil \log_2 40 \rceil$

$\lceil \log_2 2^{40} \rceil$

: 40

j = 40

A.	4
B.	5
C.	6
D.	40

40 > 1

40 > 1 − 1

20 > 1 − 2

10 > 1 − 3

5 > 1 − 4

2 > 1 − 5

1 > 1

The value returned when we call fun with the input 2^{40} is: _____

Slide

$2^{40} > 1$

$2^{39} > 1$ -

$2^{38} > 1$ -3

.

.

.

$2^{1} > 1$ —

40-1+1 = (40)

$\lfloor \log_2 2^{40} \rfloor = 40$

for(i=n; i>=2; i=i/2)

for(i=n; i>1; i=i/2)

$\lfloor \log_2 n \rfloor$

j is 40

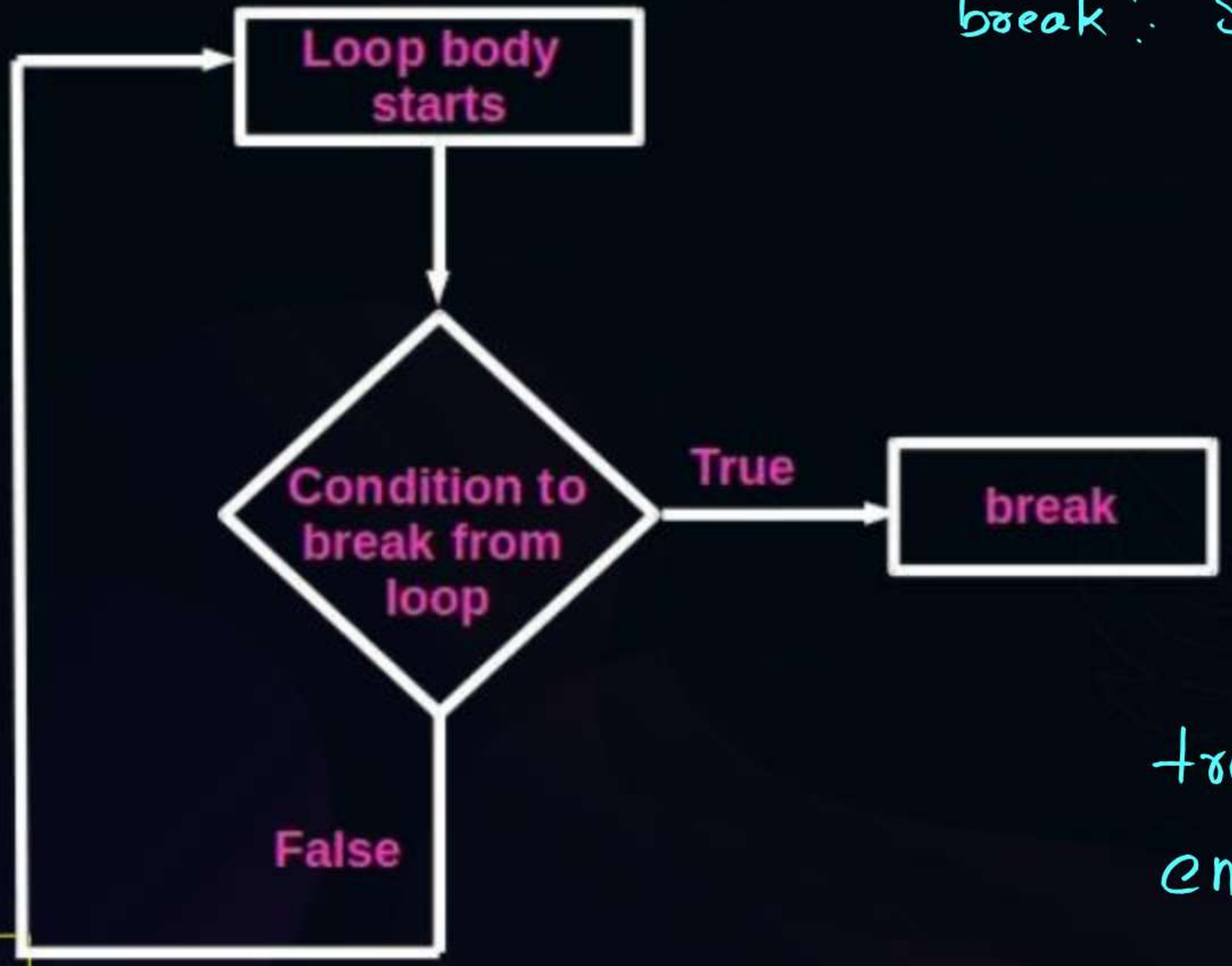
$\lfloor \log_2 40 \rfloor$ = (5)

$$for(i=n;\ i>=2;\ i=i/2)$$

$$\lfloor \log_2 n \rfloor$$



break



break : Switch statement

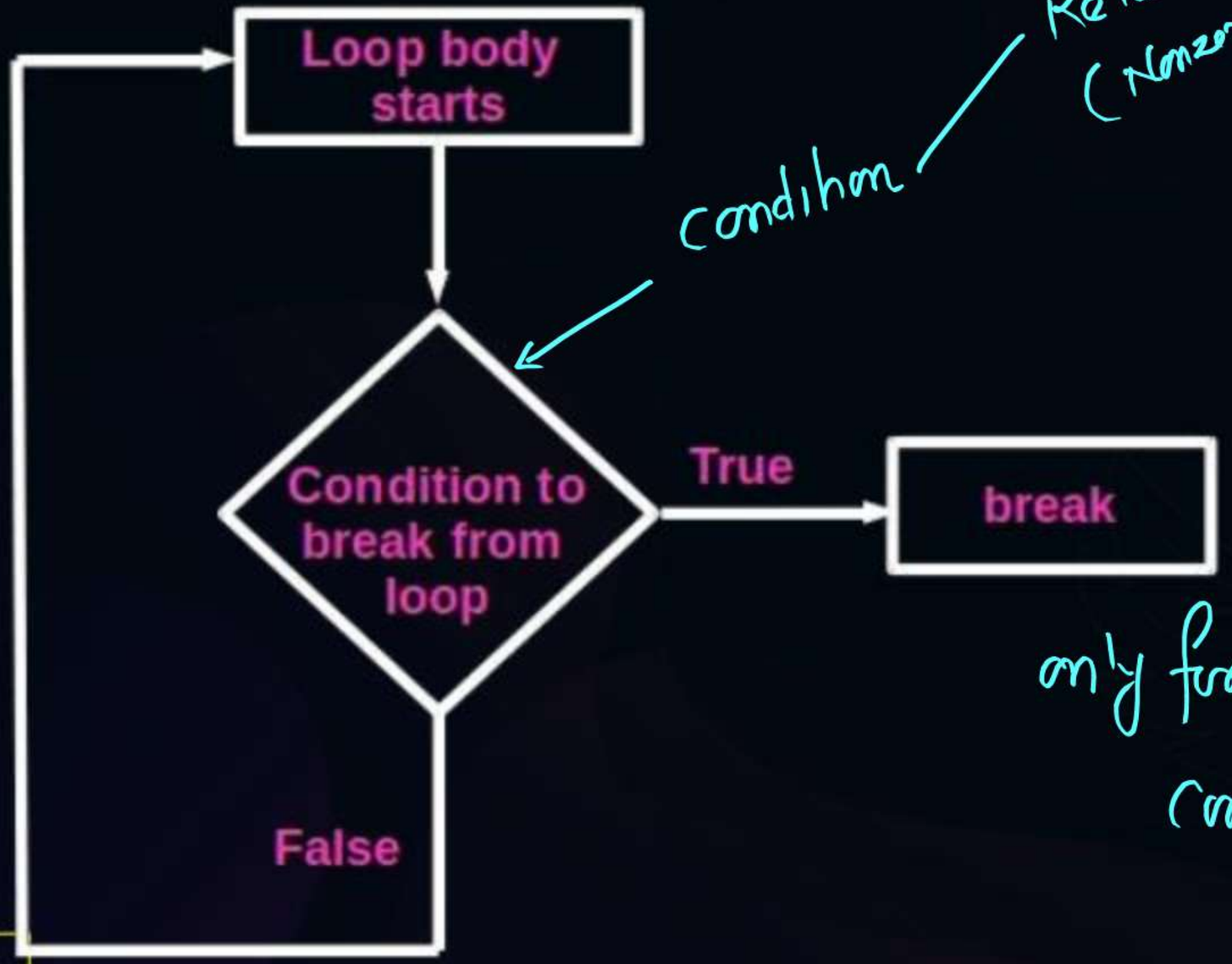
```
Switch {
case 1: stmts;
        break;
}
```

transfer the control out side of enclosed block.

Slide



break

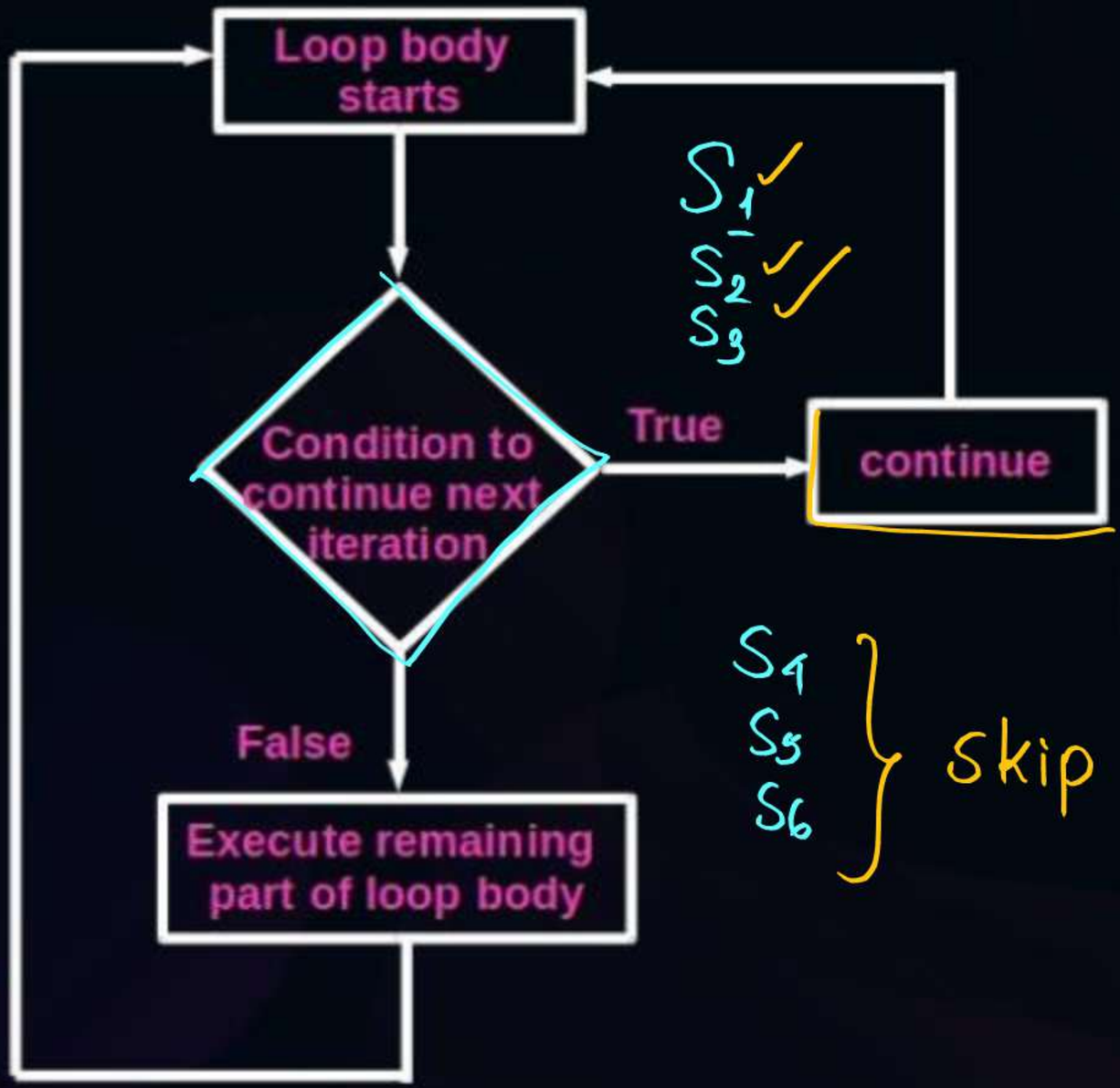


condition — Relational (Nonzero and zero)

Control will transfer out Side of enclosed loop block.

only for Immediate loop control transfer will happen.

Continue



S_1 ✓
S_2 ✓
S_3

S_4
S_5
S_6
} skip

Continue cause skip remainder body of loop and retest the condition before reiterating.

Slide



Question: Continue

#Q.

```
#include<stdio.h>
void main () {
for (int i = 1; i <= 10; i++) {
      if (i > 5) {
            break;
      }
      printf("%d\n", i);
      }
}
```

Number of times print statement executed is

A. 4

B. 5

C. 7

d. 11

break : does not take control outside if

loop

i=1	i=2	i=3	i=4	i=5
1>5	2>5	3>5	4>5	5>5
①	②	③	④	⑤

i=6

6>5

X

Slide



Question: Continue

#Q.
```
#include <stdio.h>
int main() {
    int i,j;
    int count=0;
    for(i =1;i<=3;i++){
       for(j=1;j<=20;j++){
           printf("I am a good student");
          if(j==2) break;
      }

    }
     return 0 ;
}
```

The number of times printf will be executed is _____.

i=1 — j=1 — print
j=2 — print

i=2 — j=1 — print
j=2 — print

i=3 — j=1 — print
j=2 — print

Slide



Question: Continue

#Q.

```
#include <stdio.h>
int main() {
    int i,j;
   //int count=0;
    for(i =1;i<=3;i++){
        for(j=1;j<=20;j++){
            printf("I am a good student");
           if(i==2) break;
     }

    }
     return 0 ;
}
```

(A) 40

(B) 60

(C) 41

(D) 57

20+1+20

=41

The number of times printf will be executed is [C] .

Slide

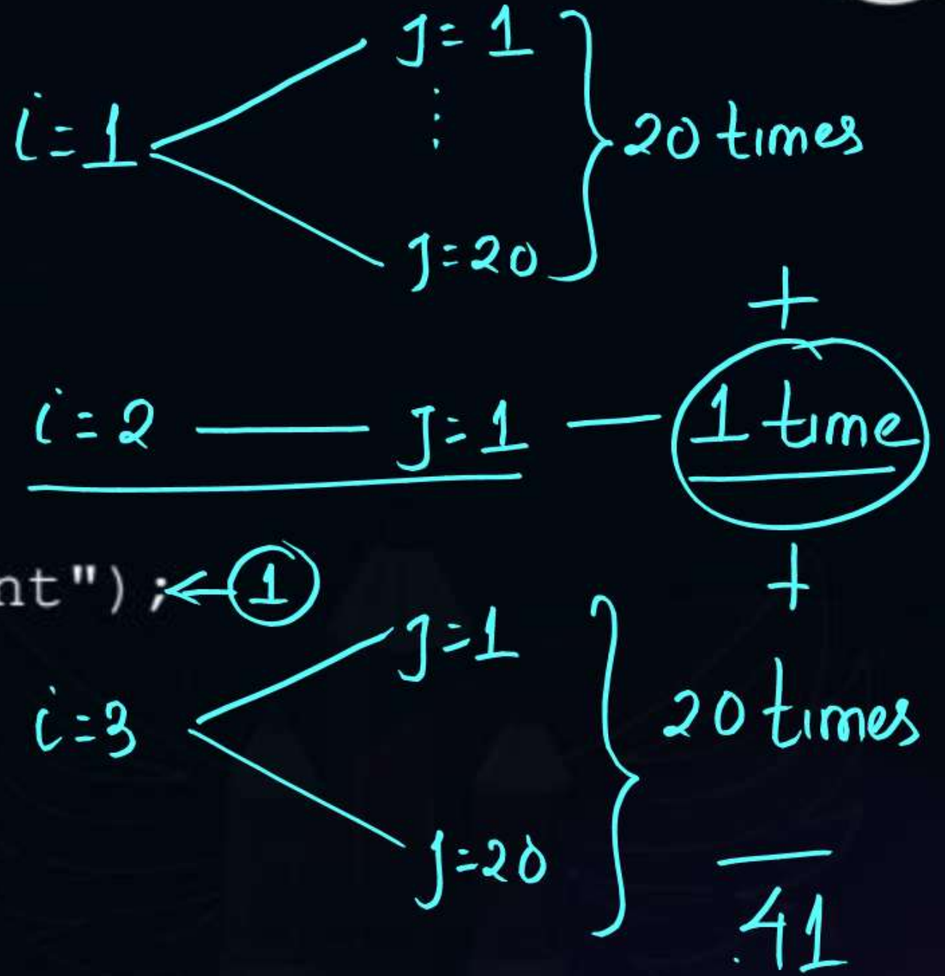


Question: Continue

#Q.
```
#include <stdio.h>
int main() {
    int i,j;
   //int count=0;
    for(i =1;i<=3;i++){
        for(j=1;j<=20;j++){
            printf("I am a good student");
           if(i==2) break;
     }

    }
     return 0 ;
}
```

The number of times printf will be executed is _____.



Slide



Question: Continue

#Q.

```
#include<stdio.h>
void main () {
for (int i = 1; i <= 10; i++) {
if (i == 5||i==6) {
      continue;
}
      printf("%d\n", i);
}
```

A 4

b.5

C.8 ✓

d. 11

No. of times printf will execute ——?

i will be Incoemont

i = 1 ✓ (1)

i = 2 ✓ (2)

i = 3 ✓ (3)

i = 4 ✓ (4)

i = 5 X

i = 6 X

i = 7 —— (5)

i = 8 —— (6)

i = 9 —— (7)

i = 10 —— (8)

Slide



Question: Continue

#Q.

```
#include<stdio.h>
void main () {
int sum =0;
for (int j =1; j<=20; j++)
 for (int i = 1; i <= 10; i++) {
     if (i == j)
          continue;
     sum++;

     }
      printf("%d\n", sum);
}
```

200
- 10
190

output of program i

j=1 — i=1,2,3..10 ——
j=2 — i=1,2,3...10 ——
j=3 — i=1,2,3..10
j=4 — i=1,2,3,4..10
j=5 — i=1,2,3,4,5 10
——

j=10 - i=1,2,3...10

j=20

value of Sum

A 200 b.190 C.180 d. 20

Slide



Question

#Q.

```
#include <stdio.h>
int main() {
    int i=2+4%6*2+9/10;
    while (i<20){
        printf("I am good student");
        i++;
        if(i%2) continue;

    }
    return 0 ;
    }
```

$2 + 4 \times 2 + 0$

$= 10$

10 - 10 < 20
11 < 20 - ②
12 < 20
13 < 20
14 < 20
15 < 20
16 < 20
17 < 20 19 < 20 ✓
18 <

The number of times printf statement executed is

A 4
b.5
C.10
d. 11

Slide

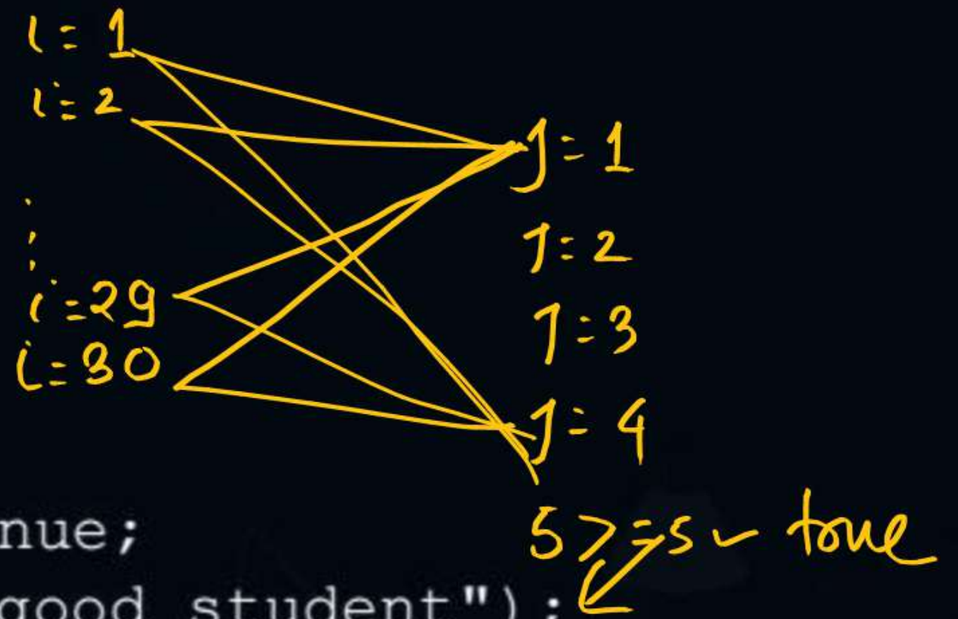
Question

#Q.

```
#include <stdio.h>
int main() {
    int i,j;

    for(i =1;i<=100;i++){
        for(j=1;j<=20;j++){
            if(j>=5) continue;
            printf("\nI am a good student");

        }
        if (i>=30) break;
    }
     return 0 ;
}
```



Slide

The number of times printf executed is _____



Question

#Q.

```
#include <stdio.h>
int main() {
      int i=10;
      for (; i<=100; i++);{
            printf("\nI am a good Student");
      }
      printf("%d",i);
      return 0 ;
      }
```

The value of I is x and number of print statement is y then value of x+y is

Slide



Question

#Q.

```
#include<stdio.h>
int main()
{
   int i=0;
   for(i=2; i<=n; i*=2)
   printf("GATE WALLAH 2024");
   return 0;
}
```

The number of times printf() executed is approximated as-

(a) 2^n (b) $\log_2 n$

(c) $\log_2 \log_2 n$ (d) 2

Slide



Question

```
#include <stdio.h>
int main() {
    int i,j=10, k=10;
        for(i=10;i<=200; i++){
            printf("\nLine 1");
        }
        do{
            printf("\nline 2");
            j = j+2;
        }while(j<=200);

       while(k<=200){
            printf("\nline 3");
            k = k+3;     }
         return 0 ;
```

The number of times line 1 printed is x , number of time line 2 printed is y and number of times line 3 printed is z then which of the following is TRUE ?

(A)x=y=z

(B)x>y>z

(C)x<y<z

(D)x=y<z

Slide



Question

#Q. The number of times loop executed is

```
#include<stdio.h>
int main(){
int i,count=0;
for (i=1; i<=100; i=i*2 ){
     printf("\nI am a good Student");

}
return 0 ;
}
```

A. 6
B. 7
C. 8
D. 10

The number of times value will be printed?

Slide



2 mins Summary

Topic — Continue

Topic — break

Topic — practic problem

Topic

Topic

THANK - YOU

